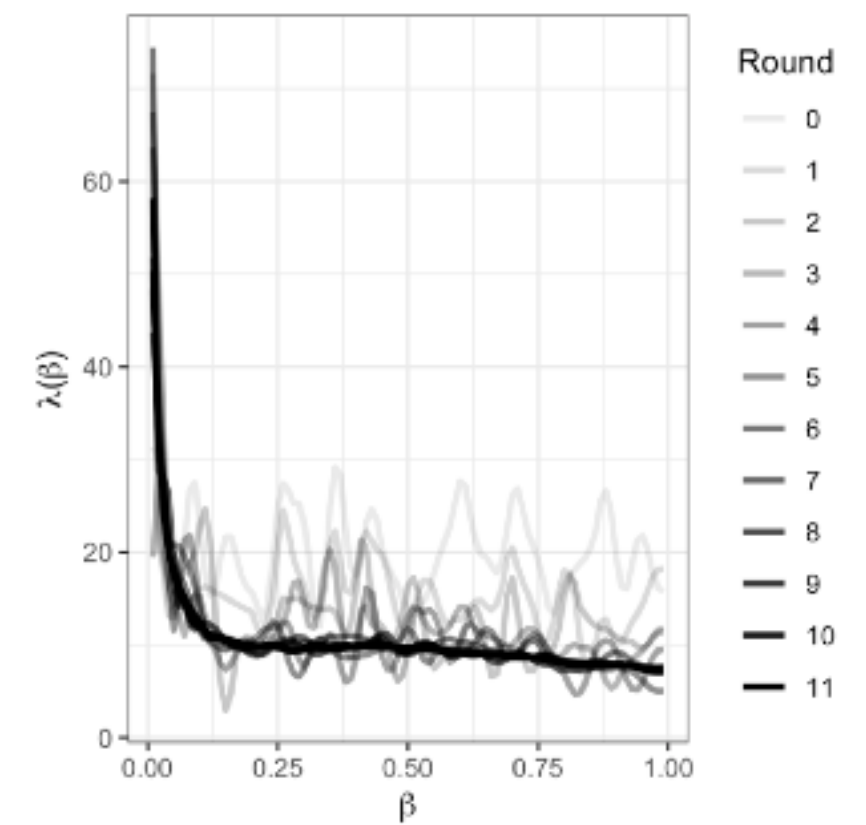
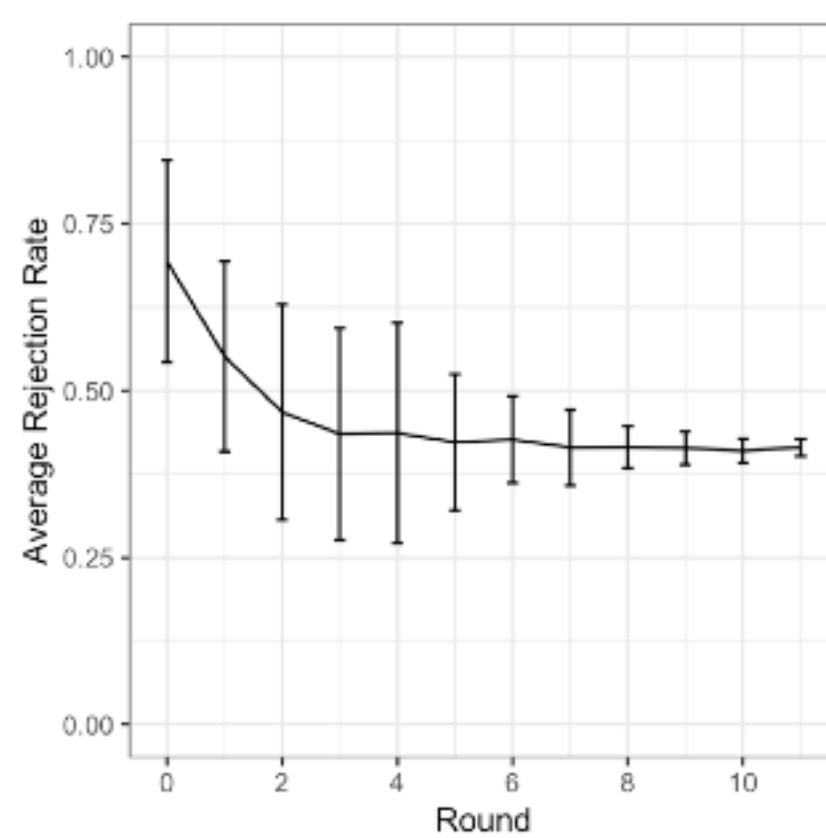
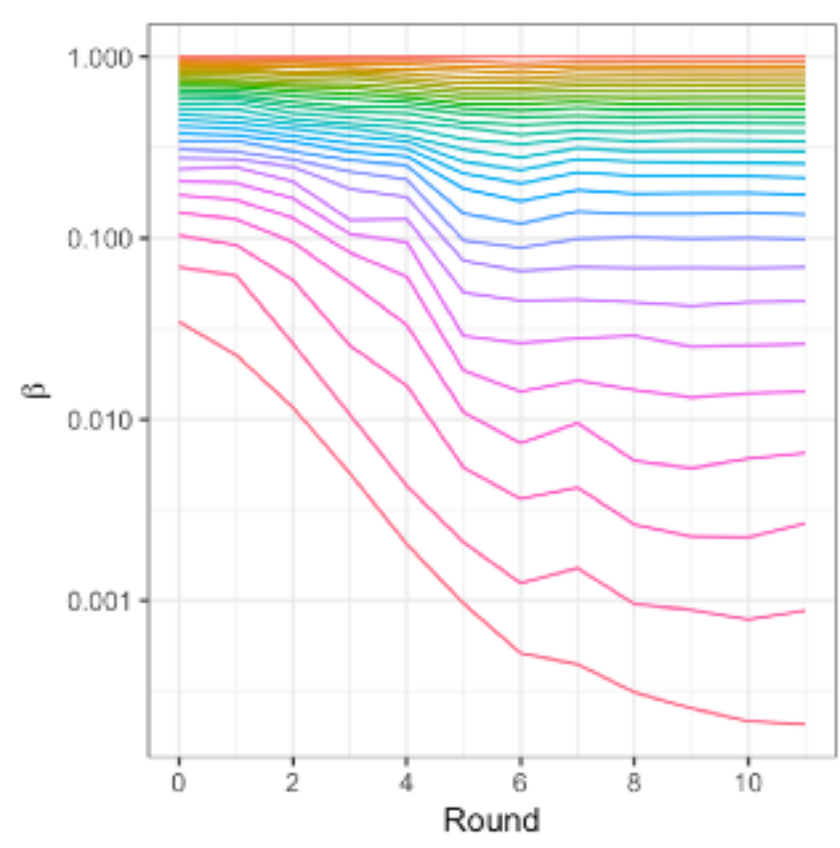


EXAMPLE: HIERARCHICAL BAYESIAN MODEL ($D = 369$)



NRPT IN PRACTICE

Model (and dataset when applicable)	n	d	$\hat{\Lambda}$	N
<i>Change point</i> detection (text message data, [Davidson-Pilon, 2015])	74	3	4.0	20
<i>Copy number</i> inference (whole genome ovarian cancer data, Section 7.5)	6 206	30	13.0	50
<i>Discrete</i> multimodal distribution (Appendix I.4.3)	N/A	3	0.4	30
Weakly identifiable <i>elliptic</i> curve (Appendix I.4)	N/A	2	4.4	30
General Linear Model (<i>GLM</i>) (Challenger O-ring dataset, [Dalal et al., 1989])	23	2	3.3	15
Bayesian <i>hierarchical</i> model (historical rocket failure data, Appendix I.4)	5 667	369	12.0	30
<i>Ising</i> model (Appendix I.4.2)	N/A	25	3.1	30
<i>Ising</i> model with <i>magnetic</i> field (Appendix I.4.2)	N/A	25	2.3	30
Bayesian <i>mixture</i> model (Section I.4.4)	300	305	8.2	30
Bayesian <i>mixture</i> model (subset of 150 datapoints)	150	155	5.5	20
<i>Isotropic normal</i> distribution	N/A	5	1.4	30
<i>ODE</i> parameters (mRNA data, [Leonhardt et al., 2014])	52	5	6.4	50
<i>Phylogenetic</i> inference (single cell breast cancer data, [Dorri et al., 2020])	192 763	192 765	88	300
<i>Phylogenetic species</i> tree inference (mtDNA, [Hayasaka et al., 1988])	249	10 395	7.1	30
Unidentifiable <i>product</i> parameterization (Appendix I.4)	100 000	2	3.7	15
<i>Rotor</i> (XY) model, [Hsieh et al., 2013]	N/A	25	3.3	40
<i>Spike-and-slab</i> classification (RMS Titanic passengers data [Hind, 2019])	200	19	4.7	30